Assignment 4: Global Supply Chain Management

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Instructions:

- As you make each decision, use the sheets below to jot down a brief record of your reasons for the choices you made each year.
- **Please upload these record sheets to the course website after completing the simulation. Thank you!**

Record Sheet, Year 1

As you make each decision, jot down a brief record of your reasons for the choices you made each year.

Design Room: Which options did you choose this year?

Option	Chosen?	Rationale for Selection
1. Upgraded Comm.	N	Standard deviation goes too high, demand goes lower
2. Exterior	N	Standard deviation goes too high, demand goes lower
3. Stylish	Y	Give higher profit, but the standard deviation does not go high
4. Storage Capacity	Y	Give higher profit, but the standard deviation does not go high

Forecast Room: What was your base forecast (mean or consensus) for each model? Did you deviate from it in your production order? If so, how? Justify both decisions.

Model	Fcst	Rationale for Forecast
Model A	62	Average "plus" standard deviation, since the markdown price is much higher than holding cost
Model B	33	A bit higher than average, since the markdown price is much higher than holding cost, but not add exactly standard deviation since the uncertainty is higher

Production Room: Which suppliers did you choose this year? Why or why not?

Supplier	Chosen?	Rationale for Selection
1. FarFarAway	Y	Lower the unit cost and set up cost
2. FarAway	N	No need to shorten the lead time with extra 1M setup cost
3. PrettyClose	Y	Get the flexibility to adjust change on demand, also lower set up cost
4 VeryClose	N	No need for higher capacity with extra 1M setup cost

How did you allocate production for the two models between your suppliers, and why?

Supplier	Model A Production	Model B Production	Month Production Started	Changes/Other decisions
1. FarFarAway	50	10	January	Change in July, 46K/10K → to match the change in demand
2. FarAway				
3. PrettyClose	12	23	May	Change in July, 0K/24K → to match the change in demand
4 VeryClose				

Did you issue a production change order this year? Did you invest in the CELLDEX show? Why or why not?

1. Yes, because we need to follow the trend of change in demand
2. No. Since we have flexibility in production, we can adjust the supplier in time and no need to pay for the CELLDEX

Boardroom Results: Did you receive a vote from the board member? What was your annual net profit?

Annual Net Profit:	$ 48,349,000

Board Member	Vote?	Comments?
Carla	Y	I think you have identified an important concept about the problems with consensus forecasts. I will give you my board vote this year. Congratulations.

Ankit	Y	I think you are right in looking at the disagreement among forecasters, since wide disagreement is often an indication of demand uncertainty. I will give you my board vote this year. Congratulations.
Mia	Y	I think you did a good job this year. I will give you my board vote. Your choice regarding the selection of suppliers at the beginning of the year have paid off. Congratulations.
Matheo	Y	You have my vote this year. You seem to have a good intuition about the differences between Model A and Model B and where to allocate production foreach model based on their differences.
Adele	Y	Congratulations on your performance this year. I will give you my vote of confidence. You seem to have a good sense of when to overproduce and when to underproduce.

Other comments and/or learnings moving forward:

Record Sheet, Year 2

As you make each decision, jot down a brief record of your reasons for the choices you made each year.

Design Room: Which options did you choose this year?

Option	Chosen?	Rationale for Selection
1. Stylish	Y	Give higher profit, but the standard deviation does not go high
2. Storage Capacity	Y	Give higher profit, but the standard deviation does not go high
3. Extended Battery	N	standard deviation goes too high, demand goes lower
4. Durability	N	standard deviation goes too high, demand goes lower

Forecast Room: What was your base forecast (mean or consensus) for each model? Did you deviate from it in your production order? If so, how? Justify both decisions.

Model	Fcst	Rationale for Forecast
Model A	57	Average "plus" standard deviation, since the markdown price is much higher than holding cost
Model B	36	A bit higher than average, since the markdown price is much higher than holding cost, but not add exactly standard deviation since the uncertainty is higher

Production Room: Which suppliers did you choose this year? Why or why not?

Supplier	Chosen?	Rationale for Selection
1. FarFarAway	Y	Lower the unit cost
2. FarAway	N	No need to shorten the lead time with extra 1M setup cost
3. PrettyClose	Y	Get the flexibility to adjust change on demand
4 VeryClose	N	No need for higher capacity with extra 1M setup cost

How did you allocate production for the two models between your suppliers, and why?

Supplier	Model A Production	Model B Production	Month Production Started	Changes/Other decisions
1. FarFarAway	50	10	January	-
2. FarAway				
3. PrettyClose	7	26	May	Change in Jul, 16K/10K → to match the change in demand
4 VeryClose				

Did you issue a production change order this year? Did you invest in the CELLDEX show? Why or why not?

1. Yes, because we need to follow the trend of change in demand
2. No. Since we have flexibility in production, we can adjust the supplier in time and no need to pay for the CELLDEX

Boardroom Results: Did you receive a vote from the board member? What was your annual net profit?

Annual Net Profit:	$53,724,120

Board Member	Vote?	Comments?
Carla	Y	You get my vote. Congratulations. For the second year in a row, I think you have been doing a great job evaluating options based on more than the consensus forecast.
Ankit	Y	Congratulations. You get my vote. I think your choices in regards to which option to choose are consistently well thought out, You seem to understand

		that the options with high demand uncertainty make accurate demand forecasting difficult.
Mia	N	I will have to withhold my vote. I am not sure you have a well aligned sourcing and production scheduling strategy yet. I would encourage you to think more about how capacity allocation and production flexibility are connected.
Matheo	Y	I will give you my vote this year. You seem to understand the difference between overseas and domestic suppliers, as well as why it is to our advantage to manufacture mostly overseas for one model but not for the other.
Adele	Y	I will give you my vote of confidence. Congratulations again on your performance this year. You seem to have picked up on the right way to look at the markdown, inventory holding, and stockout costs for each model.

Other comments and/or learnings moving forward:

Record Sheet, Year 3

As you make each decision, jot down a brief record of your reasons for the choices you made each year.

Design Room: Which options did you choose this year?

Option	Chosen?	Rationale for Selection
1. Extended Battery	N	It has no impact on profitability
2. Durability	N	Quite high SD and no higher profit
3. Anti-Theft	N	Not profitable due to its impact and SD
4. High Quality Audio	Y	Estimated monthly impact on Demand and lower SD

Forecast Room: What was your base forecast (mean or consensus) for each model? Did you deviate from it in your production order? If so, how? Justify both decisions.

Model	Fcst	Rationale for Forecast
Model A	65	Average "plus" standard deviation, since the markdown price is much higher than holding cost
Model B	30	A bit higher than average, since the markdown price is much higher than holding cost, but not add exactly standard deviation since the uncertainty is higher

Production Room: Which suppliers did you choose this year? Why or why not?

Supplier	Chosen?	Rationale for Selection
1. FarFarAway	Y	Lower the unit cost
2. FarAway	N	No need to shorten the lead time with extra 1M setup cost
3. PrettyClose	Y	Get the flexibility to adjust change on demand
4 VeryClose	N	No need for higher capacity with extra 1M setup cost

How did you allocate production for the two models between your suppliers, and why?

Supplier	Model A Production	Model B Production	Month Production Started	Changes/Other decisions
1. FarFarAway	50	10	January	-
2. FarAway				
3. PrettyClose	15	20	May	Change on Jul, 13K/14K → to match the change in demand
4 VeryClose				

Did you issue a production change order this year? Did you invest in the CELLDEX show? Why or why not?

1. Yes, because we need to follow the trend of change in demand
2. No. Since we have flexibility in production, we can adjust the supplier in time and no need to pay for the CELLDEX

Boardroom Results: Did you receive a vote from the board member? What was your annual net profit?

Annual Net Profit:	$45,725,000

Board Member	Vote?	Comments?
Carla	Y	You get my vote. Congratulations. Just like last year, you have been doing a great job on evaluating options based on more than the consensus data. You seem to understand which forecasting data are most relevant in predicting the future demand and profitability of an option. I vote yes.
Ankit	Y	Congratulations. My vote is behind you. I think your choices in regards to which option to choose are consistently well thought out, You seem to

		understand that the predictability of demand and profitability is negatively influenced by choosing certain risky options.
Mia	Y	Congratulations, your selections this year convinced me to give you my board vote. I feel like you clearly understand the importance of retaining production flexibility when setting up a production schedule. If you can keep thinking about ways to increase production flexibility, you will do fine next year as well. Remember that the challenge is to have the flexibility to adjust demand pattern safter real demand is observed, while keeping costs in mind.
Matheo	Y	I will give you my vote. You are continuing to do a good job, You seem to have understood the difference between the models when it comes to overseas versus domestic production. If you can act on your beliefs about the difference inflexibility between these sourcing possibilities, you will do fine in the next year
Adele	Y	I think you are onto an important concept of supply chain management here. Excess inventory costs (or markdown costs) and the costs of stocking out of a model are part of your calculations. Congratulations, you get my vote.

Other comments and/or learnings moving forward:

Record Sheet, Year 4

As you make each decision, jot down a brief record of your reasons for the choices you made each year.

Design Room: Which options did you choose this year?

Option	Chosen?	Rationale for Selection
1. Anti-Theft	N	Quite high sd and no higher profit
2. High Quality Audio	Y	It generates the most profit and lower the sd
3. Super-slim	N	Quite high sd and lower the average demand
4. Camera Quality	N	Quite high sd and lower the average demand

Forecast Room: What was your base forecast (mean or consensus) for each model? Did you deviate from it in your production order? If so, how? Justify both decisions.

Model	Fcst	Rationale for Forecast
Model A	64	The A is more stable, so we estimated a value=Average + sd(63+1)
Model B	30	The B is more unstable, so we have a conservative value=Average + half sd(26+4)

Production Room: Which suppliers did you choose this year? Why or why not?

Supplier	Chosen?	Rationale for Selection
1. FarFarAway	Y	Lower the unit cost
2. FarAway	N	No need to shorten the lead time with extra 1M setup cost
3. PrettyClose	Y	Get the flexibility to adjust change on demand
4 VeryClose	N	No need for higher capacity with extra 1M setup cost

How did you allocate production for the two models between your suppliers, and why?

Supplier	Model A Production	Model B Production	Month Production Started	Changes/Other decisions
1. FarFarAway	45	15	January	-
2. FarAway				
3. PrettyClose	20	15	May	Change on Jul, 5K/24K → because the news said that the sales of model A decrease, and the sales of model B is estimated to increase. News suggested that model A tend to maintain sales and model B is estimated to increase more.
4 VeryClose				

Did you issue a production change order this year? Did you invest in the CELLDEX show? Why or why not?

1. Yes, because we need to follow the trend of change in demand
2. No. Since we have flexibility in production, we can adjust the supplier in time and no need to pay for the CELLDEX

Boardroom Results: Did you receive a vote from the board member? What was your annual net profit?

Annual Net Profit:	$53,204,800

Board Member	Vote?	Comments?
Carla	Y	You get my vote this year. Congratulations. I think that you have been doing a great job on evaluating options based on more than the consensus data. You seem to understand that the average forecasting data are often more relevant in predicting the future demand and profitability of an option than

		the consensus forecasts, consensus forecasts can often be biased by group dynamics
Ankit	Y	Congratulations. My vote is behind you. I think your choices in regards to which option to choose have been consistently well thought out. You seem to understand that a high degree of variance among the different forecasters is a good indication of demand uncertainty, This uncertainty in the option can reduce the demand certainty for an otherwise certain product and, thus, profitability
Mia	Y	You have again done a great job and will receive my vote. I feel like you clearly understand the importance of increased flexibility in order to adjust to emerging demand patterns. You have attained this flexibility by splitting up the production volume between local and overseas suppliers in a suitable way.
Matheo	Y	I will give you my vote. You are continuing to do a good job. You seem to have understood the difference between Model A and B when it comes to overseas versus domestic production. As you must well understand, the models with a higher degree of certainty in demand can be scheduled more safely overseas, since they will likely need less production adjustments. The higher uncertainty models, on the other hand, need to be scheduled with the more responsive domestic suppliers.
Adele	N	I regret to not be able to give you my vote this year. I feel like you did not fully grasp the implications of stockout, holding, and markdown costs. If stockout costs (in this case the profit margins) are high relative to the markdown costs, then it is often advisable to plan for a small surplus over expected demand. If stockout costs are low relative to the cost of selling the item at a loss at the end of its life-cycle, it is often the right thing to underproduce.

Other comments and/or learnings moving forward:

Congratulations! You made it to the end of your four-year contract. I speak for the team when I say it's been wonderful working with you.